

SAT Practice Test 10

147 Questions | 3 Modules | Original Content

MODULE 1 (49 Questions)

READING & WRITING: MODULE 1

Questions 1-27 | 32 Minutes

Question 1

Skill: Words in Context | Difficulty: Medium

To preserve the fragile fibers, the conservators decided to eschew water based cleaning and instead used a dry, reversible technique.

As used in the text, what does the word "eschew" most nearly mean?

- A) embrace
- B) classify
- C) postpone
- D) deliberately avoid

Question 2

Skill: Words in Context | Difficulty: Medium

The survey participants were unaware that the researchers were measuring how often people checked the time, reducing the chance that participants would deliberately alter that behavior.

As used in the text, what does the word "unaware" most nearly mean?

- A) not knowing
- B) publicly opposed
- C) deliberately skeptical
- D) fully prepared

Question 3

Skill: Words in Context | Difficulty: Medium

The editor removed several inane remarks that sounded clever but contributed no evidence or reasoning to the argument.

As used in the text, what does the word "inane" most nearly mean?

- A) silly and lacking sense
- B) morally urgent
- C) highly technical
- D) carefully balanced

Question 4

Skill: Words in Context | Difficulty: Medium

The sculptor's idiosyncratic method of pressing woven cloth into wet clay produced textures that are immediately recognizable in her work.

As used in the text, what does the word "idiosyncratic" most nearly mean?

- A) widely standardized
- B) distinctively unusual
- C) historically inaccurate
- D) temporarily hidden

Question 5

Skill: Main Idea | Difficulty: Medium

Researchers studying salt marsh grasses have often focused on sediment retention. A recent field study found that variation in root density predicted differences in sediment retention even after the researchers accounted for site size and seasonal conditions. The finding suggests that a feature once treated as secondary may help explain why otherwise similar sites show different outcomes.

Which choice best states the main idea of the text?

- A) Scientists can now predict sediment retention without collecting field data.
- B) Variation in root density may be an important contributor to differences in sediment retention.
- C) Researchers have shown that root density is the only factor that affects sediment retention.
- D) Seasonal conditions have no measurable relationship with sediment retention.

Question 6

Skill: Main Purpose | Difficulty: Medium

To study desert beetles, researchers developed a method that measures surface groove density without disturbing the study sites. Earlier methods required removing samples, which could alter later measurements of fog-water collection. The newer approach lets researchers return to the same locations repeatedly and compare changes over time.

Which choice best states the main purpose of the text?

- A) To describe a methodological improvement and explain why it is useful for repeated observations
- B) To argue that all earlier research on the subject should be discarded
- C) To compare two competing explanations for a biological process
- D) To demonstrate that repeated measurements always produce identical results

Question 7

Skill: Overall Structure | Difficulty: Medium

For years, researchers assumed that signal transmission through traffic noise in urban songbirds was driven mainly by climate. New measurements, however, show that sites with similar climates can differ sharply in song frequency. The researchers therefore propose that future models include both climate and song frequency rather than treating climate as sufficient by itself.

Which choice best describes the overall structure of the text?

- A) It presents two equally supported theories and argues that no choice between them is possible.
- B) It presents a conventional explanation, introduces evidence that complicates that explanation, and states a revised approach.
- C) It describes an experiment, rejects its results, and proposes repeating it with a larger sample.
- D) It lists several unrelated observations and ranks them by importance.

Question 8

Skill: Detail (According to the Text) | Difficulty: Medium

A study of mangrove seedlings compared 21 study units with higher values of tidal water exchange and 29 study units with lower values. Researchers measured seedling survival at each study unit on 5 occasions under comparable procedures. They found that units with higher tidal water exchange had a larger average value of seedling survival, but the difference between the groups was smallest on the final observation.

According to the text, which statement is true?

- A) The difference in seedling survival between the two groups was smallest on the final observation.
- B) Exactly 50 study units had higher values of tidal water exchange.
- C) Sites with lower tidal water exchange had the larger average value of seedling survival on every occasion.
- D) The study units were randomly assigned to the higher and lower groups.

Question 9

Skill: Inference / Complete the Text | Difficulty: Medium

Researchers studying coral reef fish found that observations with higher algal grazing intensity also tended to show greater coral recruitment. During a follow-up period, naturally occurring changes in algal grazing intensity were followed by changes in coral recruitment in the same direction, while observations with little change in algal grazing intensity showed little change in the outcome. The researchers did not test every possible background factor.

Which choice most logically follows from the text?

- A) The results prove that algal grazing intensity is the only cause of variation in coral recruitment.
- B) Because not every background factor was tested, the observed pattern provides no information about the relationship.
- C) The results are consistent with a relationship in which changes in algal grazing intensity may contribute to changes in coral recruitment, though alternative explanations remain possible.
- D) Any observation with low algal grazing intensity must also have low coral recruitment.

Question 10

Skill: Evidence - Data / Graph / Table | Difficulty: Medium

Researchers studying coastal dunes predicted that sites with more deep-rooted grass cover would tend to have greater sand retention. The table shows measurements from four study units collected using the same procedure.

Group	Deep-Rooted Grass Cover	Sand Retention
A	16	31
B	19	38
C	22	47
D	25	55

Which choice most effectively uses data from the table to support the researchers' prediction?

- A) The value of sand retention decreased at every site as deep-rooted grass cover increased.
- B) Groups B and C had identical values for both variables.
- C) Group D had the highest measured deep-rooted grass cover and the highest measured sand retention.
- D) Group A had the lowest measured deep-rooted grass cover but the highest measured sand retention.

Question 11

Skill: Evidence - Support / Weaken | Difficulty: Medium

Researchers hypothesize that contextual predictability directly increases word-recognition speed in bilingual readers, rather than merely being correlated with another favorable background condition.

Which finding, if true, would most strongly support the hypothesis?

- A) Researchers use several instruments to measure word-recognition speed.
- B) In a controlled experiment, researchers increased contextual predictability for one group while holding other measured conditions constant; word-recognition speed increased relative to the control group.
- C) Some sites with low contextual predictability nevertheless show moderate word-recognition speed.
- D) Sites with high contextual predictability are more common in regions where bilingual readers have been studied for many years.

Question 12

Skill: Cross-Text | Difficulty: Medium

Text 1: A researcher argues that differences in surface cooling among city trees are best explained by canopy cover. Because the variables track one another across many sites, the researcher recommends treating canopy cover as the primary indicator in regional models.

Text 2: Another researcher agrees that canopy cover can be informative but notes that the relationship weakens in extreme environments. This researcher recommends combining canopy cover with a second indicator rather than relying on it alone.

Based on the texts, how would the author of Text 2 most likely respond to the claim in Text 1?

- A) Text 2 would claim that regional models should exclude environmental information entirely.
- B) Text 2 would argue that the indicator has no relationship to the outcome under any conditions.
- C) Text 2 would agree that the proposed indicator is useful but would reject treating it as sufficient in all conditions.
- D) Text 2 would agree that the indicator is always the primary cause of the outcome.

Question 13

Skill: Function (Underlined Portion) | Difficulty: Medium

Researchers studying forest fungi initially used hyphal-network connectivity as a proxy for nutrient transfer among plants. That choice made fieldwork faster, but it also meant that any factor affecting the proxy independently could distort the estimates. A later study therefore measured both variables directly at a subset of sites.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It states the later study's main numerical result.
- B) It identifies both an advantage and a limitation of the earlier method, motivating the later study.
- C) It argues that direct measurement is always faster than using a proxy.
- D) It provides historical background unrelated to the change in method.

Question 14

Skill: Quotation Illustrates Claim | Difficulty: Medium

In an original fictional passage, Evan is in a winter garden. The narrator portrays Evan as experiencing hope tempered by awareness of fragility: _____

Which quotation most effectively illustrates the claim?

- A) "Evan saw the green shoots and became certain that nothing could harm the garden now."
- B) "Evan ignored the new shoots because he expected every plant to fail."
- C) "The glass roof of the winter garden collected beads of condensation."
- D) "Evan smiled at the first green shoots rising through the soil, then covered them carefully, knowing that one hard frost could undo the promise they seemed to carry."

Question 15

Skill: Grammar / Conventions | Difficulty: Medium

A research team labeled this analysis Study 10. The density of microscopic pores, along with the thickness of the outer coating, _____ how quickly water vapor leaves sample 4.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) determine
- B) have determined
- C) are determining
- D) determines

Question 16

Skill: Grammar / Conventions | Difficulty: Medium

A research team labeled this analysis Study 11. The first survey recorded only daytime _____ the second survey included both daytime and nighttime traffic.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) traffic; the
- B) traffic: and the
- C) traffic the
- D) traffic, the

Question 17

Skill: Grammar / Conventions | Difficulty: Medium

A research team labeled this analysis Study 12. Built from strips of cedar bark and dyed fiber, _____ preserved in the museum is unusually flexible despite its age.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) its flexibility was
- B) the museum preserved the basket
- C) the basket
- D) researchers found the basket

Question 18

Skill: Grammar / Conventions | Difficulty: Medium

A research team labeled this analysis Study 13. The new sensors are more precise than _____ used in the 1990s, especially when temperatures change rapidly.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) it
- B) these one
- C) that
- D) those

Question 19

Skill: Grammar / Conventions | Difficulty: Medium

A research team labeled this analysis Study 14. During the field season, the researchers sampled river water, measured sediment depth, and _____ the location of each observation.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) records
- B) to record
- C) recording
- D) recorded

Question 20

Skill: Grammar / Conventions | Difficulty: Medium

A research team labeled this analysis Study 15. By the time the second expedition reached the valley, the first team _____ its equipment and left the site.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) will pack
- B) has packed
- C) had packed
- D) packs

Question 21

Skill: Transitions | Difficulty: Medium

Earlier surveys treated the two pottery styles as contemporaneous. _____ radiocarbon dates from sealed layers place one style at least a century earlier.

Which choice completes the text with the most logical transition?

- A) However,
- B) For example,
- C) Similarly,
- D) Therefore,

Question 22

Skill: Transitions | Difficulty: Medium

The coastal population declined after a severe storm. _____ the inland population, which experienced the same temperature change but little storm damage, remained stable.

Which choice completes the text with the most logical transition?

- A) For instance,
- B) Consequently,
- C) By contrast,
- D) In other words,

Question 23

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

1. Topic: Maritime trade in Southeast Asia and Europe
2. Southeast Asian ports often linked island and mainland trade through monsoon routes
3. Many medieval European ports were connected by shorter seasonal routes in enclosed seas
4. Historian L. Arif argues that geography shaped different scheduling strategies in the two regions

The student wants to compare how geography influenced trade organization in Southeast Asia and Europe. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Many medieval European ports were connected by shorter seasonal routes in enclosed seas
- B) Historian L. Arif argues that geography shaped different scheduling strategies in the two regions
- C) Southeast Asian ports often depended on monsoon routes linking islands and mainlands, whereas many medieval European ports operated along shorter seasonal routes in enclosed seas, illustrating how geography shaped different trade schedules in the two regions.
- D) Southeast Asian ports often linked island and mainland trade through monsoon routes

Question 24

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- 1. Topic: Henrietta Leavitt and Cepheid variables
- 2. Henrietta Leavitt studied variable stars at the Harvard College Observatory
- 3. She discovered a relationship between a Cepheid variable star's period and luminosity
- 4. Astronomers later used that relationship to estimate vast cosmic distances

The student wants to explain why Leavitt's discovery became important to astronomy. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Henrietta Leavitt studied variable stars at the Harvard College Observatory
- B) She discovered a relationship between a Cepheid variable star's period and luminosity
- C) Astronomers later used that relationship to estimate vast cosmic distances
- D) Leavitt discovered a relationship between Cepheid variables' periods and luminosities, a finding that later allowed astronomers to estimate vast cosmic distances.

Question 25

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- 1. Topic: Mangrove restoration
- 2. A restoration team planted the same mangrove species at 12 coastal sites
- 3. Seedling survival was highest near tidal channels with regular water exchange
- 4. Sites farther from channels had saltier soils and lower survival

The student wants to recommend where future plantings should be concentrated. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Sites farther from channels had saltier soils and lower survival
- B) Seedling survival was highest near tidal channels with regular water exchange
- C) A restoration team planted the same mangrove species at 12 coastal sites
- D) Because seedling survival was highest near tidal channels with regular water exchange and lower at saltier sites farther away, future mangrove plantings should be concentrated near well-flushed tidal channels.

Question 26

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- 1. Topic: Alma Thomas

2. Alma Thomas was an American painter associated with abstract color painting
3. In 1972 she became the first Black woman to receive a solo exhibition at the Whitney Museum of American Art
4. Her paintings often use repeated blocks of vivid color

The student wants to identify a major milestone in Thomas's career. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Alma Thomas was an American painter associated with abstract color painting
- B) In 1972, Alma Thomas became the first Black woman to receive a solo exhibition at the Whitney Museum of American Art, a major milestone in her career.
- C) Her paintings often use repeated blocks of vivid color
- D) Alma Thomas is associated with several developments described in the notes.

Question 27

Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

1. Topic: Battery cooling systems
2. Liquid cooling reduced peak battery temperature by 18 percent in a vehicle test
3. The cooling system added mass and consumed some electrical power
4. Engineers evaluate thermal performance together with energy and weight costs

The student wants to present a tradeoff rather than a one-sided benefit. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Liquid cooling reduced peak battery temperature by 18 percent, but it also added mass and consumed electrical power, so its thermal benefit must be weighed against energy and weight costs.
- B) The cooling system added mass and consumed some electrical power
- C) Engineers evaluate thermal performance together with energy and weight costs
- D) Liquid cooling reduced peak battery temperature by 18 percent in a vehicle test

MATH: MODULE 1

Questions 28-49 | 35 Minutes

You may use a calculator for all questions in this section.

Question 28

Skill: Linear Equations in One & Two Variables | Difficulty: Medium

$$3x + 11 = 32$$

What is the value of x ?

- A) 21
- B) 9
- C) 7
- D) 5

Question 29

Skill: Percentages | Difficulty: Medium

What is 35% of 240?

- A) 85

- B) 35
- C) 84
- D) 156

Question 30

Skill: Ratios, Rates & Proportions | Difficulty: Medium

The ratio of red objects to blue objects is 3:5. There are 48 objects in total.

How many of the objects are in the first group?

- A) 8
- B) 18
- C) 45
- D) 30

Question 31

Skill: Function Notation & Evaluation | Difficulty: Medium

$$f(x) = 4x + 7$$

What is the value of $f(5)$?

- A) 27
- B) 12
- C) 11
- D) 20

Question 32

Skill: Coordinate Geometry | Difficulty: Medium

The line passes through (2, 4) and (6, 12).

What is the slope of the line through the two points?

- A) -2
- B) 0.5
- C) 4
- D) 2

Question 33

Skill: Systems of Linear Equations | Difficulty: Medium

$$2x + y = 11$$

$$x - y = 1$$

What is the solution to the system?

- A) $x = 4, y = 3$
- B) $x = 5, y = 2$
- C) $x = 6, y = 1$
- D) $x = 3, y = 5$

Question 34

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Medium

$$(4x^3)(5x^4)$$

Which expression is equivalent to the product?

- A) $20x^4$
- B) $20x^7$
- C) $20x^{12}$
- D) $9x^7$

Question 35

Skill: Statistics: Mean & Median | Difficulty: Medium

The mean of five numbers is 13. Four of the numbers are 8, 11, 14, and 17.

What is the fifth number?

- A) 15
- B) 12
- C) 13
- D) 20

Question 36

Skill: Area & Perimeter | Difficulty: Medium

A rectangle has length 3 times its width and perimeter 56.

What is the width of the rectangle?

- A) 21
- B) 28
- C) 7
- D) 14

Question 37

Skill: Probability - Basic | Difficulty: Medium

A box contains 25 items, of which 7 are marked. One item is selected at random.

What is the probability of selecting a marked item at random?

- A) $7/18$
- B) $1/7$
- C) $18/25$
- D) $7/25$

Question 38

Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

A quantity starts at 200 and is multiplied by 1.2 each year. What is its value after 3 years?

- A) 320
- B) 345.6
- C) 720
- D) 288

Question 39

Skill: Quadratic Equations - Solving | Difficulty: Medium

$$x^2 - 10x + 21 = 0$$

What are the solutions to the equation?

- A) 3 and 7
- B) -3 and -7
- C) 10 and 21
- D) 3 only

Question 40

Skill: Circle Equations | Difficulty: Medium

$$(x - 2)^2 + (y + 3)^2 = 25$$

What is the radius of the circle?

- A) 10
- B) 25
- C) 5
- D) 6

Question 41

Skill: Unit Conversion & Dimensional Analysis | Difficulty: Medium

A length is 48 meters. Use 1 meter = 100 centimeters.

How many centimeters is this length?

- A) 480
- B) 48000
- C) 48
- D) 4800

Question 42

Skill: Linear Inequalities | Difficulty: Medium

$$3x + 5 < 29$$

Which inequality is equivalent?

- A) $x > 9$
- B) $x < 8$
- C) $x > 8$
- D) $x < 24$

Question 43

Skill: Similar Triangles | Difficulty: Medium

Two triangles are similar. A side of length 4 in the smaller triangle corresponds to a side of length 12 in the larger triangle. Another side of the smaller triangle has length 10.

What is the corresponding side length in the larger triangle?

- A) 30

- B) 90
- C) 13
- D) 3

Question 44

Skill: Interpreting Coefficients in Linear Models | Difficulty: Medium

A linear model is $y = 6x + 40$.

What does the coefficient of x represent in context?

- A) x increases by 40 whenever y increases by 1.
- B) The predicted value decreases by 40 for each 1-unit increase in x .
- C) The predicted value increases by 6 for each 1-unit increase in x .
- D) The predicted value is 6 when $x = 0$.

Question 45

Skill: Polynomial Operations | Difficulty: Medium

$(2x + 3) + (4x - 5)$

Which expression is equivalent?

- A) $4x$
- B) $6x + 8$
- C) $8x - 2$
- D) $6x - 2$

Question 46

Skill: Volume & Surface Area | Difficulty: Medium

A cylinder has radius 3 and height 8. Express the answer in terms of π .

What is the volume of the cylinder?

- A) 17π
- B) 24π
- C) 48π
- D) 72π

Question 47

Skill: Statistical Inference & Sampling | Difficulty: Medium

A researcher estimates student exercise habits by surveying only students who voluntarily attend an after-school fitness club.

Which concern most directly limits the conclusion?

- A) The sample is too large to analyze.
- B) The sample may not represent all students at the school.
- C) Every student in the school had an equal chance of selection.
- D) Exercise habits cannot be measured with surveys.

Question 48

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Medium

If $5x + 9 = 39$, what is the value of $10x + 18$?

Student-Produced Response

Enter your answer in the response field.

Question 49

Skill: Systems of Linear Equations | Difficulty: Medium

A store packs items in boxes of 6 and boxes of 11. A customer buys exactly 5 boxes containing exactly 45 items. How many boxes of 6 were purchased?

Student-Produced Response

Enter your answer in the response field.

MODULE 2: EASIER ADAPTIVE PATH (49 Questions)

READING & WRITING: EASIER MODULE

Questions 50-76 | 32 Minutes

Question 50

Skill: Words in Context | Difficulty: Easy

Reviewers often extol the architect's use of daylight, praising the way narrow courtyards illuminate rooms without excessive heat.

As used in the text, what does the word "extol" most nearly mean?

- A) question cautiously
- B) imitate exactly
- C) reduce sharply
- D) praise highly

Question 51

Skill: Words in Context | Difficulty: Easy

The apparent correlation was spurious: once the researchers controlled for age, the relationship disappeared.

As used in the text, what does the word "spurious" most nearly mean?

- A) partly complete
- B) well established
- C) unexpectedly useful
- D) false or not genuine

Question 52

Skill: Words in Context | Difficulty: Easy

A chamber group preparing for a festival selected forty pieces that its musicians could perform without additional rehearsal. The group expanded this repertory over the following year as it learned new works.

As used in the text, what does the word "repertory" most nearly mean?

- A) stock of works or skills available for performance
- B) audience response to a performance
- C) chronological record of rehearsals
- D) method of preserving documents

Question 53

Skill: Words in Context | Difficulty: Easy

The finding was counterintuitive: adding a second exit slowed evacuation because people crowded toward the more visible doorway.

As used in the text, what does the word "counterintuitive" most nearly mean?

- A) impossible to test
- B) consistent across all cases
- C) too obvious to dispute
- D) contrary to what one would expect

Question 54

Skill: Main Idea | Difficulty: Easy

Researchers studying city trees have often focused on surface cooling. A recent field study found that variation in canopy cover predicted differences in surface cooling even after the researchers accounted for site size and seasonal conditions. The finding suggests that a feature once treated as secondary may help explain why otherwise similar sites show different outcomes.

Which choice best states the main idea of the text?

- A) Researchers have shown that canopy cover is the only factor that affects surface cooling.
- B) Scientists can now predict surface cooling without collecting field data.
- C) Variation in canopy cover may be an important contributor to differences in surface cooling.
- D) Seasonal conditions have no measurable relationship with surface cooling.

Question 55

Skill: Main Purpose | Difficulty: Easy

To study forest fungi, researchers developed a method that measures hyphal-network connectivity without disturbing the study sites. Earlier methods required removing samples, which could alter later measurements of nutrient transfer among plants. The newer approach lets researchers return to the same locations repeatedly and compare changes over time.

Which choice best states the main purpose of the text?

- A) To argue that all earlier research on the subject should be discarded
- B) To describe a methodological improvement and explain why it is useful for repeated observations
- C) To compare two competing explanations for a biological process
- D) To demonstrate that repeated measurements always produce identical results

Question 56

Skill: Overall Structure | Difficulty: Easy

For years, researchers assumed that ultraviolet-damage resistance in alpine flowers was driven mainly by climate. New measurements, however, show that sites with similar climates can differ sharply in protective pigment concentration. The researchers therefore propose that future models include both climate and protective pigment concentration rather than treating climate as sufficient by itself.

Which choice best describes the overall structure of the text?

- A) It presents a conventional explanation, introduces evidence that complicates that explanation, and states a revised approach.
- B) It presents two equally supported theories and argues that no choice between them is possible.
- C) It lists several unrelated observations and ranks them by importance.
- D) It describes an experiment, rejects its results, and proposes repeating it with a larger sample.

Question 57

Skill: Detail (According to the Text) | Difficulty: Easy

A study of sea turtles compared 18 study units with higher values of access to magnetic-field cues and 26 study units with lower values. Researchers measured navigation accuracy at each study unit on 2 occasions under comparable procedures. They found that units with higher access to magnetic-field cues had a larger average value of navigation accuracy, but the difference between the groups was smallest on the final observation.

According to the text, which statement is true?

- A) Exactly 44 study units had higher values of access to magnetic-field cues.

- B) The difference in navigation accuracy between the two groups was smallest on the final observation.
- C) Sites with lower access to magnetic-field cues had the larger average value of navigation accuracy on every occasion.
- D) The study units were randomly assigned to the higher and lower groups.

Question 58

Skill: Inference / Complete the Text | Difficulty: Easy

Researchers studying battery cells found that observations with higher separator porosity also tended to show greater heat dissipation. During a follow-up period, naturally occurring changes in separator porosity were followed by changes in heat dissipation in the same direction, while observations with little change in separator porosity showed little change in the outcome. The researchers did not test every possible background factor.

Which choice most logically follows from the text?

- A) Because not every background factor was tested, the observed pattern provides no information about the relationship.
- B) The results are consistent with a relationship in which changes in separator porosity may contribute to changes in heat dissipation, though alternative explanations remain possible.
- C) The results prove that separator porosity is the only cause of variation in heat dissipation.
- D) Any observation with low separator porosity must also have low heat dissipation.

Question 59

Skill: Inference / Complete the Text | Difficulty: Easy

Researchers studying prairie seeds found that observations with higher brief heat exposure also tended to show greater germination after fire. During a follow-up period, naturally occurring changes in brief heat exposure were followed by changes in germination after fire in the same direction, while observations with little change in brief heat exposure showed little change in the outcome. The researchers did not test every possible background factor.

Which choice most logically follows from the text?

- A) Because not every background factor was tested, the observed pattern provides no information about the relationship.
- B) Any observation with low brief heat exposure must also have low germination after fire.
- C) The results prove that brief heat exposure is the only cause of variation in germination after fire.
- D) The results are consistent with a relationship in which changes in brief heat exposure may contribute to changes in germination after fire, though alternative explanations remain possible.

Question 60

Skill: Evidence - Data / Graph / Table | Difficulty: Easy

Researchers studying urban foxes predicted that sites with more exposure to temporary barriers would tend to have greater route-learning speed. The table shows measurements from four study units collected using the same procedure.

Group	Exposure To Temporary Barriers	Route-Learning Speed
A	12	33
B	15	40
C	18	49
D	21	57

Which choice most effectively uses data from the table to support the researchers' prediction?

- A) Group D had the highest measured exposure to temporary barriers and the highest measured route-learning speed.
- B) Group A had the lowest measured exposure to temporary barriers but the highest measured route-learning speed.
- C) Groups B and C had identical values for both variables.
- D) The value of route-learning speed decreased at every site as exposure to temporary barriers increased.

Question 61

Skill: Evidence - Support / Weaken | Difficulty: Easy

Researchers claim that differences in large-predator abundance explain why two groups of kelp forests differ in urchin grazing pressure.

Which finding, if true, would most directly weaken the claim?

- A) A second team can reproduce the measurements of large-predator abundance.
- B) The difference in urchin grazing pressure is also observed during another month.
- C) After the groups were matched for a separate environmental factor, the difference in urchin grazing pressure disappeared even though the groups still differed in large-predator abundance.
- D) The groups differ in large-predator abundance by a statistically measurable amount.

Question 62

Skill: Cross-Text | Difficulty: Easy

Text 1: A researcher argues that differences in peak flood reduction among urban wetlands are best explained by temporary water-storage capacity. Because the variables track one another across many sites, the researcher recommends treating temporary water-storage capacity as the primary indicator in regional models.

Text 2: Another researcher agrees that temporary water-storage capacity can be informative but notes that the relationship weakens in extreme environments. This researcher recommends combining temporary water-storage capacity with a second indicator rather than relying on it alone.

Based on the texts, how would the author of Text 2 most likely respond to the claim in Text 1?

- A) Text 2 would argue that the indicator has no relationship to the outcome under any conditions.
- B) Text 2 would agree that the indicator is always the primary cause of the outcome.
- C) Text 2 would claim that regional models should exclude environmental information entirely.
- D) Text 2 would agree that the proposed indicator is useful but would reject treating it as sufficient in all conditions.

Question 63

Skill: Function (Underlined Portion) | Difficulty: Easy

Researchers studying marine sponges initially used microbial-symbiont diversity as a proxy for nutrient-cycling rate. That choice made fieldwork faster, but it also meant that any factor affecting the proxy independently could distort the estimates. A later study therefore measured both variables directly at a subset of sites.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It states the later study's main numerical result.
- B) It argues that direct measurement is always faster than using a proxy.
- C) It provides historical background unrelated to the change in method.
- D) It identifies both an advantage and a limitation of the earlier method, motivating the later study.

Question 64

Skill: Quotation Illustrates Claim | Difficulty: Easy

In an original fictional passage, Lena is in a rehearsal room. The narrator portrays Lena as experiencing confidence that grows through revision rather than sudden inspiration: _____

Which quotation most effectively illustrates the claim?

- A) "Lena declared the first draft finished because revision would only weaken her original inspiration."
- B) "Lena crossed out four measures of the unfinished score and rewrote them slowly; with each revision, the music sounded less accidental and her confidence became steadier."
- C) "Several music stands had been stacked against the rehearsal-room wall."
- D) "Lena waited for a perfect melody to arrive all at once before changing a single note."

Question 65

Skill: Grammar / Conventions | Difficulty: Easy

A research team labeled this analysis Study 10. The experiment produced one especially surprising _____ seedlings exposed to brief cold periods later tolerated heat better than untreated seedlings.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) result,
- B) result; because
- C) result:
- D) result and

Question 66

Skill: Grammar / Conventions | Difficulty: Easy

A research team labeled this analysis Study 11. The murals covering the station wall _____ scenes from the neighborhood's industrial history.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) depict
- B) depicting
- C) depicts
- D) has depicted

Question 67

Skill: Grammar / Conventions | Difficulty: Easy

A research team labeled this analysis Study 12. The team released the software under an open license, _____ independent researchers to inspect and reproduce the analysis.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) allowed
- B) allowing
- C) to have allowed
- D) allows

Question 68

Skill: Grammar / Conventions | Difficulty: Easy

A research team labeled this analysis Study 13. Neither the field notebooks nor the final report _____ an explanation for why sample 7 was excluded.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) provides
- B) were providing
- C) provide
- D) have provided

Question 69

Skill: Grammar / Conventions | Difficulty: Easy

A research team labeled this analysis Study 14. The two _____ research focused on different archives, but both studies reached similar conclusions.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) sisters's
- B) sister's
- C) sisters'
- D) sisters

Question 70

Skill: Transitions | Difficulty: Easy

The team first mapped every visible crack in the mural. _____ it compared those maps with infrared images to identify repairs hidden beneath later paint.

Which choice completes the text with the most logical transition?

- A) Nevertheless,
- B) Then,
- C) Instead,
- D) Conversely,

Question 71

Skill: Transitions | Difficulty: Easy

The sample was small and came from a single valley. _____ the consistency of the chemical pattern across all sampled sites makes the finding worth testing in a broader survey.

Which choice completes the text with the most logical transition?

- A) Nevertheless,
- B) Likewise,
- C) For example,
- D) As a result,

Question 72

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

1. Topic: Two editions of a novel
2. A novel was first published in 1818 and substantially revised in 1831
3. The later edition changes several passages about a central character's motivations
4. Literary scholars compare the editions to study how the author's ideas evolved

The student wants to explain why comparing the editions can be useful. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Literary scholars compare the editions to study how the author's ideas evolved
- B) Because the 1831 edition substantially changes passages about a central character's motivations, comparing it with the 1818 edition can reveal how the author's ideas evolved.
- C) The later edition changes several passages about a central character's motivations
- D) A novel was first published in 1818 and substantially revised in 1831

Question 73

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

1. Topic: Urban bird survey
2. Researchers counted birds in one densely built district for four weeks
3. They found more species near small parks than on blocks without vegetation
4. The district may not represent lower-density neighborhoods in the same city

The student wants to acknowledge a limitation on generalizing the study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The survey found more bird species near small parks, but because it covered only one densely built district for four weeks, its results may not generalize to lower-density neighborhoods.
- B) They found more species near small parks than on blocks without vegetation
- C) Researchers counted birds in one densely built district for four weeks They found more species near small parks than on blocks without vegetation
- D) Researchers counted birds in one densely built district for four weeks

Question 74

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

1. Topic: Great Zimbabwe
2. Great Zimbabwe includes monumental dry-stone architecture
3. Imported glass beads and ceramics have been found at the site
4. The finds indicate participation in long-distance trade networks

The student wants to connect architecture with evidence of broader economic connections. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Imported glass beads and ceramics have been found at the site
- B) The finds indicate participation in long-distance trade networks
- C) Great Zimbabwe includes monumental dry-stone architecture
- D) Great Zimbabwe's monumental dry-stone architecture and the imported glass beads and ceramics found there together show that the site combined major local building traditions with participation in long-distance trade networks.

Question 75

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

1. Topic: Indirect requests
2. A linguistics study compared requests among close friends and strangers
3. Speakers used more indirect forms with strangers in the sampled conversations
4. The researchers interpret the pattern as sensitivity to social distance

The student wants to state the study's interpretation of its main finding. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) A linguistics study compared requests among close friends and strangers
- B) Speakers used more indirect forms with strangers in the sampled conversations
- C) Because speakers used more indirect requests with strangers than with close friends, the researchers interpret indirectness as partly reflecting sensitivity to social distance.
- D) A linguistics study compared requests among close friends and strangers Speakers used more indirect forms with strangers in the sampled conversations

Question 76

Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

1. Topic: City tree irrigation
2. Newly planted street trees had 82 percent survival when watered weekly during the first summer
3. Trees watered only during visible wilting had 55 percent survival
4. The city has limited staff for watering

The student wants to recommend a policy supported by the evidence. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Newly planted street trees had 82 percent survival when watered weekly during the first summer
- B) The city has limited staff for watering
- C) Since weekly watering produced 82 percent survival compared with 55 percent for watering only after visible wilting, the city should prioritize scheduled weekly watering for newly planted trees during their first summer.
- D) Trees watered only during visible wilting had 55 percent survival

MATH: EASIER MODULE

Questions 77-98 | 35 Minutes

You may use a calculator for all questions in this section.

Question 77

Skill: Linear Equations in One & Two Variables | Difficulty: Easy

$$3x + 19 = 52$$

What is the value of x ?

- A) 11
- B) 13
- C) 33
- D) 9

Question 78

Skill: Percentages | Difficulty: Easy

What is 35% of 320?

- A) 113
- B) 35
- C) 208
- D) 112

Question 79

Skill: Ratios, Rates & Proportions | Difficulty: Easy

The ratio of red objects to blue objects is 3:5. There are 64 objects in total.

How many of the objects are in the first group?

- A) 61
- B) 24
- C) 40
- D) 8

Question 80

Skill: Function Notation & Evaluation | Difficulty: Easy

$$f(x) = 4x + 11$$

What is the value of $f(5)$?

- A) 15
- B) 20
- C) 16
- D) 31

Question 81

Skill: Coordinate Geometry | Difficulty: Easy

The line passes through (2, 8) and (6, 20).

What is the slope of the line through the two points?

- A) -3
- B) 3
- C) $\frac{1}{3}$

D) 4

Question 82

Skill: Systems of Linear Equations | Difficulty: Easy

$$2x + y = 13$$

$$x - y = -1$$

What is the solution to the system?

A) $x = 6, y = 3$

B) $x = 5, y = 4$

C) $x = 4, y = 5$

D) $x = 3, y = 7$

Question 83

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Easy

$$(4x^3)(5x^6)$$

Which expression is equivalent to the product?

A) $20x^{18}$

B) $9x^9$

C) $20x^6$

D) $20x^9$

Question 84

Skill: Statistics: Mean & Median | Difficulty: Easy

The mean of five numbers is 15. Four of the numbers are 8, 11, 14, and 17.

What is the fifth number?

A) 15

B) 30

C) 25

D) 12

Question 85

Skill: Area & Perimeter | Difficulty: Easy

A rectangle has length 3 times its width and perimeter 88.

What is the width of the rectangle?

A) 22

B) 44

C) 11

D) 33

Question 86

Skill: Probability - Basic | Difficulty: Easy

A box contains 45 items, of which 7 are marked. One item is selected at random.

What is the probability of selecting a marked item at random?

- A) $\frac{1}{7}$
- B) $\frac{38}{45}$
- C) $\frac{7}{38}$
- D) $\frac{7}{45}$

Question 87

Skill: Exponential Functions - Growth & Decay | Difficulty: Easy

A quantity starts at 200 and is multiplied by 1.1 each year. What is its value after 3 years?

- A) 260
- B) 266.2
- C) 660
- D) 242

Question 88

Skill: Quadratic Equations - Solving | Difficulty: Easy

$$x^2 - 12x + 27 = 0$$

What are the solutions to the equation?

- A) 3 and 9
- B) 3 only
- C) -3 and -9
- D) 12 and 27

Question 89

Skill: Circle Equations | Difficulty: Easy

$$(x - 2)^2 + (y + 5)^2 = 25$$

What is the radius of the circle?

- A) 5
- B) 10
- C) 25
- D) 7

Question 90

Skill: Unit Conversion & Dimensional Analysis | Difficulty: Easy

A length is 60 meters. Use 1 meter = 100 centimeters.

How many centimeters is this length?

- A) 600
- B) 6000
- C) 60000
- D) 60

Question 91

Skill: Linear Inequalities | Difficulty: Easy

$$3x + 5 < 41$$

Which inequality is equivalent?

- A) $x > 13$
- B) $x < 12$
- C) $x > 12$
- D) $x < 36$

Question 92

Skill: Similar Triangles | Difficulty: Easy

Two triangles are similar. A side of length 6 in the smaller triangle corresponds to a side of length 12 in the larger triangle. Another side of the smaller triangle has length 14.

What is the corresponding side length in the larger triangle?

- A) 56
- B) 16
- C) 28
- D) 7

Question 93

Skill: Interpreting Coefficients in Linear Models | Difficulty: Easy

A linear model is $y = 6x + 60$.

What does the coefficient of x represent in context?

- A) The predicted value increases by 6 for each 1-unit increase in x .
- B) The predicted value is 6 when $x = 0$.
- C) x increases by 60 whenever y increases by 1.
- D) The predicted value decreases by 60 for each 1-unit increase in x .

Question 94

Skill: Polynomial Operations | Difficulty: Easy

$$(2x + 3) + (4x - 9)$$

Which expression is equivalent?

- A) $8x - 6$
- B) $6x + 12$
- C) $0x$
- D) $6x - 6$

Question 95

Skill: Volume & Surface Area | Difficulty: Easy

A cylinder has radius 3 and height 12. Express the answer in terms of π .

What is the volume of the cylinder?

- A) 108π
- B) 21π
- C) 36π

D) 72π

Question 96

Skill: Statistical Inference & Sampling | Difficulty: Easy

A researcher estimates student study habits by surveying only students who voluntarily subscribe to an academic tutoring newsletter.

Which concern most directly limits the conclusion?

- A) The sample is too large to analyze.
- B) Every student in the school had an equal chance of selection.
- C) The sample may not represent all students at the school.
- D) Exercise habits cannot be measured with surveys.

Question 97

Skill: Equivalent Expressions & Algebraic Manipulation | Difficulty: Easy

If $5x + 9 = 59$, what is the value of $10x + 18$?

Student-Produced Response

Enter your answer in the response field.

Question 98

Skill: Systems of Linear Equations | Difficulty: Easy

A store packs items in boxes of 6 and boxes of 11. A customer buys exactly 7 boxes containing exactly 62 items. How many boxes of 6 were purchased?

Student-Produced Response

Enter your answer in the response field.

MODULE 2: HARDER ADAPTIVE PATH (49 Questions)

READING & WRITING: HARDER MODULE

Questions 99-125 | 32 Minutes

Question 99

Skill: Words in Context | Difficulty: Hard

The archive director gave conservators considerable latitude in choosing reversible treatments, provided that every intervention was documented and could later be undone.

As used in the text, what does the word "latitude" most nearly mean?

- A) freedom to act or make choices
- B) strict obligation
- C) geographic distance from the equator
- D) public approval

Question 100

Skill: Words in Context | Difficulty: Hard

Rather than accuse the earlier editors of suppressing evidence, the historian makes an oblique criticism, noting only that several inconvenient letters were somehow absent from the published collection.

As used in the text, what does the word "oblique" most nearly mean?

- A) physically elevated
- B) widely repeated
- C) mathematically exact
- D) indirect rather than openly stated

Question 101

Skill: Words in Context | Difficulty: Hard

By adulthood, the birds had expanded their behavioral repertory: individuals could open unfamiliar seed pods, exploit new feeding sites, and switch among several alarm calls.

As used in the text, what does the word "repertory" most nearly mean?

- A) stock of methods or works available for use
- B) sequence of historical dates
- C) formal agreement between institutions
- D) written record of ownership

Question 102

Skill: Words in Context | Difficulty: Hard

The isotope evidence is equivocal. It is consistent with seasonal migration, but the same pattern could arise if the animals remained local and shifted between foods with different chemical signatures.

As used in the text, what does the word "equivocal" most nearly mean?

- A) unrelated to the issue
- B) completely decisive
- C) deliberately fabricated

D) open to more than one interpretation

Question 103

Skill: Main Idea | Difficulty: Hard

Researchers have often treated tree-ring density as a direct proxy for summer moisture. A new multi-region study found a strong relationship under ordinary conditions, but the relationship weakened sharply where wood anatomy altered how the proxy was recorded without comparably changing the target variable. After explicitly modeling wood anatomy, the proxy again predicted summer moisture well across most regions. The authors therefore reject neither the proxy nor the complication: they argue that the proxy is informative only when the mechanism affecting its measurement is incorporated into the model.

Which choice best states the main idea of the text?

- A) tree-ring density is too unreliable to provide information about summer moisture under any conditions.
- B) The study proves that ordinary conditions are scientifically unimportant compared with extreme conditions.
- C) wood anatomy directly determines summer moisture in every region studied.
- D) tree-ring density can be a useful indicator of summer moisture, but only if researchers account for conditions in which wood anatomy changes the proxy independently of the target.

Question 104

Skill: Main Purpose | Difficulty: Hard

A common method for estimating summer moisture uses a fast indirect measurement that is inexpensive enough for large surveys. The method is usually accurate, but laboratory work shows that wood anatomy can produce the same signal even when summer moisture is unchanged. Rather than abandon the indirect measure, researchers propose taking a smaller set of direct measurements at sites where wood anatomy is likely to be unusually strong, then using those direct measurements to calibrate the larger survey.

Which choice best states the main purpose of the text?

- A) To demonstrate that calibration is unnecessary whenever a survey includes many sites
- B) To argue that indirect measurements should always be replaced by direct measurements regardless of cost
- C) To identify a condition that can make a convenient proxy misleading and explain a targeted strategy for reducing that problem without discarding the proxy
- D) To compare two unrelated research traditions and show that they reach identical conclusions

Question 105

Skill: Overall Structure | Difficulty: Hard

An influential model assumes that changes in tree-ring density and summer moisture should occur together because both respond to the same underlying process. The text first explains why the model gained support from broad regional averages. It then presents fine-scale observations in which the variables temporarily diverge. Next, it identifies wood anatomy as a mechanism capable of producing the divergence without overturning the underlying process. Finally, it argues that the original model remains useful if its predictions are stated at the appropriate spatial and temporal scale.

Which choice best describes the overall structure of the text?

- A) It presents an observation, argues that the observation is a measurement error, and recommends ignoring all fine-scale data.
- B) It explains why a model became persuasive, introduces evidence that appears to challenge it, supplies a mechanism that reconciles the evidence with a qualified version of the model, and limits the scale of the model's claims.
- C) It describes a model, proves the model universally correct, and dismisses apparent exceptions as irrelevant.

D) It lists several competing models, rejects each one, and concludes that the phenomenon cannot be studied quantitatively.

Question 106

Skill: Detail (According to the Text) | Difficulty: Hard

In a field experiment, researchers grouped sites by baseline tree-ring density but did not randomly assign sites to those groups. They then introduced the same standardized change in wood anatomy at every site. Sites with intermediate baseline values showed large changes in summer moisture; sites already near the observed upper limit of summer moisture changed little. A second analysis matched sites on four measured background variables and produced the same pattern, although the authors note that unmeasured differences could remain.

According to the text, which statement is true?

- A) The second analysis contradicted the pattern observed in the first analysis.
- B) Random assignment guarantees that no unmeasured differences existed among the baseline groups.
- C) The matched-site analysis reproduced the pattern in which sites near the upper limit of summer moisture changed less than sites with intermediate baseline values.
- D) Every site showed the same absolute change in summer moisture after the standardized change in wood anatomy.

Question 107

Skill: Inference / Complete the Text | Difficulty: Hard

A model predicts that increasing wood anatomy raises summer moisture only until another resource becomes limiting. In one experiment, wood anatomy increased substantially in all treatment groups. Summer moisture rose in low-resource groups but barely changed in high-resource groups, which already had values close to the maximum observed elsewhere. The researchers did not directly measure the proposed limiting resource.

Which choice most logically follows from the text?

- A) The results are consistent with, but do not prove, the idea that the weak response in high-resource groups reflects a ceiling or another limiting factor rather than the complete irrelevance of wood anatomy.
- B) The model requires identical absolute increases in summer moisture across all starting conditions.
- C) The experiment proves that wood anatomy has no effect on summer moisture whenever baseline values are high.
- D) Because the limiting resource was not directly measured, the increase in the low-resource groups provides no evidence relevant to the model.

Question 108

Skill: Inference / Complete the Text | Difficulty: Hard

Across cities, higher tree-ring density is associated with higher summer moisture. One explanation is causal. Another is that a third variable influences both. Researchers then examine only cities in which the proposed third variable falls within a very narrow range. The association between tree-ring density and summer moisture remains, but it is smaller than in the full sample.

Which choice most logically follows from the text?

- A) The proposed third variable cannot affect either variable because it varied little in the restricted sample.
- B) The restricted analysis proves that tree-ring density directly causes summer moisture.
- C) Because the association became smaller, the original association must have been entirely spurious.
- D) Restricting variation in the proposed third variable weakens a simple confounding explanation, but the reduced association and observational design still leave other explanations possible.

Question 109

Skill: Evidence - Data / Graph / Table | Difficulty: Hard

[GRAPH TO INSERT: line graph. X-axis: treatment intensity 0, 1, 2, 3. Y-axis: mean response. Group A values: 10, 18, 24, 27. Group B values: 10, 15, 18, 19. Error bars are similar in size.]

Researchers predicted that the response would increase with the x-variable in both groups but that the increase would be stronger in Group A. Which choice best describes the data relative to the prediction?

Which choice most effectively uses data from the graph?

- A) The data support the prediction only if Group A has a lower final value than Group B.
- B) The data show identical responses because both groups begin at the same value.
- C) The data support the prediction: both groups increase overall, and the total increase from the first to the last x-value is larger for Group A than for Group B.
- D) The data contradict the prediction because Group B never decreases between adjacent x-values.

Question 110

Skill: Evidence - Data / Graph / Table | Difficulty: Hard

[GRAPH TO INSERT: scatterplot. X-axis: pore density from 10 to 70. Y-axis: drying time from 2 to 18 hours. Nine points follow a strong downward trend from about (12,17) to (68,4). One point at approximately (60,15) lies far above the trend line.]

A researcher claims that the graph supports a negative association but also suggests that an additional variable may strongly affect at least one observation. Which feature of the graph most directly supports the second part of the claim?

Which feature most directly supports the claim?

- A) The fact that most y-values are positive
- B) The observation at a high x-value that lies far above the otherwise strong downward trend
- C) The fact that the x-axis begins above zero
- D) The presence of several observations near the trend line

Question 111

Skill: Evidence - Support / Weaken | Difficulty: Hard

Researchers hypothesize that increasing wood anatomy directly changes summer moisture, rather than merely tracking another favorable condition.

Which finding, if true, would most strongly support the hypothesis?

- A) Some sites with low wood anatomy nevertheless have moderate values of summer moisture.
- B) In a randomized experiment, researchers manipulated wood anatomy while holding measured background conditions constant, and the treatment group showed the predicted change in summer moisture relative to a control group.
- C) A second instrument ranks sites by wood anatomy almost exactly as the first instrument does.
- D) Across a larger observational sample, wood anatomy and summer moisture remain correlated, though less strongly than before.

Question 112

Skill: Evidence - Support / Weaken | Difficulty: Hard

Researchers claim that differences in tree-ring density explain why two populations differ in summer moisture.

Which finding, if true, would most directly weaken the claim?

- A) A second measure of tree-ring density produces nearly the same ranking as the first.
- B) The difference in summer moisture is observed again in another season.

- C) Across additional sites, tree-ring density and summer moisture remain correlated, though the association is somewhat weaker.
- D) After the populations are matched on a separate environmental factor, the difference in summer moisture disappears even though the populations still differ substantially in tree-ring density.

Question 113

Skill: Cross-Text | Difficulty: Hard

Text 1: A researcher argues that tree-ring density is a robust regional indicator of summer moisture because the two variables are strongly associated across a large pooled data set. The researcher recommends using one calibration equation across all regions.

Text 2: A second researcher finds that the association is strong within each region but that the intercept differs systematically among regions because of wood anatomy. Pooling the regions therefore hides meaningful baseline differences, even though the within-region relationship is real.

Based on the texts, how would the author of Text 2 most likely respond to the recommendation in Text 1?

- A) By agreeing that the indicator can be informative while arguing that a single pooled calibration may be misleading unless regional baseline differences are modeled
- B) By concluding that the indicator should be discarded because regional intercepts differ
- C) By denying that any relationship exists within individual regions
- D) By arguing that pooling data necessarily makes every estimate more accurate

Question 114

Skill: Function (Underlined Portion) | Difficulty: Hard

A broad survey found that tree-ring density tracked summer moisture closely in most sites. The exceptions were concentrated almost entirely in sites where wood anatomy changed abruptly during the measurement interval. The researchers argue that this clustering matters because a random measurement problem would not be expected to occur specifically under those conditions. They therefore propose that the proxy responds more slowly than the target variable when conditions change quickly.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It characterizes the exceptions in a nonrandom pattern that supports the researchers' proposed explanation for when the proxy fails.
- B) It dismisses the exceptional sites as irrelevant because they are few in number.
- C) It proves that all measurement error is caused by abrupt environmental change.
- D) It introduces an unrelated observation solely to broaden the subject of the passage.

Question 115

Skill: Quotation Illustrates Claim | Difficulty: Hard

In an original fictional passage, the narrator presents Leena's response as confidence in a method tempered by awareness of what the method cannot resolve.

Which quotation most effectively illustrates the claim?

- A) Leena copied the first number she saw and left before the instrument finished calibrating.
- B) Leena decided that uncertainty made measurement pointless and refused to record any result.
- C) Leena checked the calibration twice, pleased that the readings agreed to the third decimal place, then paused over the final uncertainty range, which no additional adjustment could make disappear.
- D) Leena admired the instrument because its brass case matched the laboratory shelves.

Question 116

Skill: Grammar / Conventions | Difficulty: Hard

Neither the unusually high values recorded at the coastal sites nor the single inland outlier _____ sufficient, by itself, to establish that the instrument was miscalibrated.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) have been
- B) is
- C) were
- D) are

Question 117

Skill: Grammar / Conventions | Difficulty: Hard

The researchers expected the proxy to fail under rapid _____ the direct measurements, however, showed that the largest errors occurred after conditions had already stabilized.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) change the
- B) change, the
- C) change; the
- D) change: although the

Question 118

Skill: Grammar / Conventions | Difficulty: Hard

Based on measurements collected at twelve sites, _____ initially appeared to support the regional model.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) the researchers concluded that the pooled pattern
- B) there was a pooled pattern that researchers
- C) the pattern in the pooled data
- D) the regional model was what the researchers supported

Question 119

Skill: Grammar / Conventions | Difficulty: Hard

The archive contains three versions of the map, each with annotations in a different hand; determining which annotations are contemporaneous with the original map _____ additional chemical analysis.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) requires
- B) have required
- C) require
- D) are requiring

Question 120

Skill: Grammar / Conventions | Difficulty: Hard

By the time the laboratory adopted the revised protocol, researchers at two other institutions _____ the same source of contamination independently.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) identified
- B) had identified
- C) have identified
- D) will identify

Question 121

Skill: Transitions | Difficulty: Hard

The researchers wanted to determine whether the proxy failed specifically during abrupt environmental change. _____ they installed direct sensors at sites expected to experience rapid transitions and collected measurements at much shorter intervals.

Which choice completes the text with the most logical transition?

- A) To this end,
- B) Nevertheless,
- C) Similarly,
- D) In contrast,

Question 122

Skill: Transitions | Difficulty: Hard

The pooled data suggest a strong relationship between the variables. _____ within each region, the slopes are similar but the intercepts differ substantially, meaning the pooled equation masks systematic regional baselines.

Which choice completes the text with the most logical transition?

- A) For example,
- B) However,
- C) Likewise,
- D) Accordingly,

Question 123

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching the topic described in the notes below, a student has taken notes about: A historian compares port records from 1710 and 1730.

1. A historian compares port records from 1710 and 1730.
2. The number of recorded arrivals rose by 40%.
3. A new customs office opened in 1722 and began recording small coastal vessels that had often gone uncounted.
4. The student wants to describe the increase without assuming that all of it reflects a true increase in shipping.

Which choice most effectively uses relevant information from the notes to accomplish the stated goal?

- A) A new customs office opened in 1722 and began recording small coastal vessels that had often gone uncounted.
- B) The number of recorded arrivals rose by 40%.
- C) Recorded arrivals were 40% higher in 1730 than in 1710, but because a new customs office began counting small coastal vessels that had often gone unrecorded, the increase cannot automatically be interpreted as a 40% rise in actual shipping activity.
- D) The number of recorded arrivals rose by 40%. A new customs office opened in 1722 and began recording small coastal vessels that had often gone uncounted.

Question 124

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching the topic described in the notes below, a student has taken notes about: A botanist compares seed size across two islands.

1. A botanist compares seed size across two islands.
2. Island A has larger average seeds.
3. Bird species that disperse the seeds also differ between the islands.
4. The student wants to state a plausible explanation without claiming causation.

Which choice most effectively uses relevant information from the notes to accomplish the stated goal?

- A) Island A has larger average seeds. Bird species that disperse the seeds also differ between the islands.
- B) Bird species that disperse the seeds also differ between the islands.
- C) Island A has larger average seeds.
- D) Seeds are larger on average on Island A, and differences in seed-dispersing bird communities offer one plausible explanation, though the comparison alone does not establish that the birds caused the difference.

Question 125

Skill: Rhetorical Synthesis | Difficulty: Hard

While researching the topic described in the notes below, a student has taken notes about: A study tests a new reading intervention.

1. A study tests a new reading intervention.
2. Students were assigned by existing classroom, not individually randomized.
3. Intervention classrooms improved more than comparison classrooms.
4. The student wants to report the result while acknowledging the design limitation.

Which choice most effectively uses relevant information from the notes to accomplish the stated goal?

- A) Students were assigned by existing classroom, not individually randomized.
- B) Students in intervention classrooms improved more than those in comparison classrooms, but because assignment occurred by existing classroom rather than by individual randomization, preexisting classroom differences remain a possible explanation for part of the effect.
- C) Students were assigned by existing classroom, not individually randomized. Intervention classrooms improved more than comparison classrooms.
- D) Intervention classrooms improved more than comparison classrooms.

MATH: HARDER MODULE

Questions 126-147 | 35 Minutes

This module is intentionally written at the upper end of official Digital SAT difficulty while remaining within tested SAT content.

Question 126

Skill: Linear & Exponential Models from Data | Difficulty: Hard

[GRAPH TO INSERT: scatterplot. X-axis x and y-axis y . A line of best fit is drawn with equation $y = 6x - 17$. The plotted points cluster closely around the line; exact point coordinates are not needed.]

New variables are defined by $u = (x - 40)/5$ and $v = 2y + 9$. Which equation represents the same line of best fit in terms of u and v ?

Which equation is correct?

- A) $v = 12u + 455$
- B) $v = 30u + 223$

C) $v = 60u + 455$

D) $v = 60u - 25$

Question 127

Skill: Volume & Surface Area | Difficulty: Hard

Solid B is similar to solid A. The volume of B is $125/27$ times the volume of A. The surface area of A is 270. What is the surface area of B?

What is the surface area of B?

A) 450

B) 750

C) 1250

D) $486/5$

Question 128

Skill: Factoring Polynomials | Difficulty: Hard

$(x^2 + ax + 13)(x^2 - ax + 13) = x^4 - 10x^2 + 169$, where a is positive. What is $a + 13$?

What is $a + 13$?

A) 30

B) 49

C) 7

D) 19

Question 129

Skill: Circle Equations | Difficulty: Hard

A circle has center $(2, -3)$ and passes through $P(8, 5)$. The line tangent to the circle at P intersects the vertical line $x = 10$ at Q . What is the y -coordinate of Q ?

What is the y -coordinate of Q ?

A) -3

B) $7/2$

C) $3/4$

D) 5

Question 130

Skill: Circles: Arc Length & Sector Area | Difficulty: Hard

From a point outside a circle, two tangent segments touch the circle at A and B and form an angle of 46 degrees. An inscribed angle has endpoints A and B and its vertex lies on the minor arc AB , so it intercepts the major arc AB . What is the measure, in degrees, of the inscribed angle?

What is the angle measure?

A) 113

B) 67

C) 134

D) 46

Question 131

Skill: Percentages | Difficulty: Hard

After increasing by 12%, a price is reduced by 15% of the increased price. The final price is 476.00 dollars. What was the original price, in dollars?

What was the original price?

- A) 500
- B) 476
- C) 425
- D) 560

Question 132

Skill: Absolute Value Equations | Difficulty: Hard

What is the sum of all real solutions to $|3x - 8| = 2x + 7$?

What is the sum of the solutions?

- A) 15
- B) $16/3$
- C) $1/5$
- D) $76/5$

Question 133

Skill: Radical and Rational Exponents | Difficulty: Hard

The equation $\sqrt{2x + 3} = x - 3$ has exactly one real solution. What is that solution?

What is the solution?

- A) $\sqrt{40}$
- B) 8
- C) $4 + \sqrt{10}$
- D) $4 - \sqrt{10}$

Question 134

Skill: Nonlinear Systems of Equations | Difficulty: Hard

The graphs $y = x^2 - 6x + 10$ and $y = 2x + k$ intersect at exactly one point. What is k ?

What is the value of k ?

- A) 16
- B) -6
- C) 10
- D) -10

Question 135

Skill: Discriminant & Number of Solutions | Difficulty: Hard

The equation $(k + 2)x^2 + 2(k - 5)x + (k - 8) = 0$ has exactly one real solution in x . What is k ?

What is the value of k ?

- A) $-41/4$

- B) $\frac{41}{4}$
- C) $\frac{45}{4}$
- D) $\frac{37}{4}$

Question 136

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Hard

The parabola $y = x^2 - 8x + c$ and the horizontal line $y = 5$ are tangent at exactly one point, directly above $x = 4$. What is the value of c ?

What is c ?

- A) 21
- B) 16
- C) 23
- D) 5

Question 137

Skill: Similar Triangles | Difficulty: Hard

Two similar triangles have areas that differ by 96 square units. The smaller triangle has area 54 square units. A side of the smaller triangle is 9. What is the corresponding side length in the larger triangle?

What is the corresponding side length?

- A) 15
- B) 27
- C) 45
- D) 14

Question 138

Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

[GRAPH TO INSERT: exponential-function graph. The graph of $y = f(x)$ passes through (2, 18) and (5, 486) and is exponential with a constant multiplicative factor for each 1-unit increase in x .]

What is $f(0)$?

What is $f(0)$?

- A) 18
- B) 3
- C) 2
- D) 162

Question 139

Skill: Function Notation & Evaluation | Difficulty: Hard

$f(x) = 3x - 2$ and $g(x) = 2x + 5$. If $f(g(k)) = 55$, what is $g(k) + k$?

What is $g(k) + k$?

- A) 19
- B) 55
- C) 7
- D) 26

Question 140

Skill: Rational Expressions & Equations | Difficulty: Hard

For x not equal to 3, solve $(x + 7)/(x - 3) = 15/(x - 3) + 2$. What is x ?

What is x ?

- A) 7
- B) -3
- C) 3
- D) -2

Question 141

Skill: Statistical Inference & Sampling | Difficulty: Hard

A school randomly assigns 200 volunteers to two study methods and finds a significant difference in test scores. The volunteers were recruited from an advanced-placement review course, but the school wants to generalize the result to all students in the district.

Which statement best evaluates the study?

- A) Because the participants were randomly assigned, the results can automatically be generalized to every member of the target population.
- B) Because participation was voluntary, no causal conclusion can be drawn even within the studied sample.
- C) Random assignment supports a causal comparison within the studied participants, but the restricted volunteer sample limits how confidently the result can be generalized to the broader target population.
- D) A larger sample would eliminate any concern about how the participants were recruited.

Question 142

Skill: Standard Deviation | Difficulty: Hard

Data set A has standard deviation 6. Data set B is formed by replacing every value a with $-3a + 12$. Data set C is formed by replacing every value b in B with $(b - 5)/2$. What is the standard deviation of C?

What is the standard deviation of C?

- A) 9
- B) 12.5
- C) 3
- D) 18

Question 143

Skill: Conditional Probability & Two-Way Tables | Difficulty: Hard

At a school of 300 students, $2/5$ participate in a tutoring club. Of club participants, $3/4$ passed an exam; of nonparticipants, $1/3$ passed. If a randomly selected student is known to have passed, what is the probability that the student participated in the club?

What is the probability?

- A) $120/300$
- B) $90/300$
- C) $60/150$
- D) $3/5$

Question 144

Skill: Quadratic Functions - Graphs & Properties | Difficulty: Hard

[GRAPH TO INSERT: parabola. The graph crosses the x-axis at $x = 2$ and $x = 10$. Its vertex is at $(6, 32)$ and the parabola opens downward.]

If the graph is $y = a(x - r_1)(x - r_2)$, what is a ?

What is a ?

- A) -32
- B) 32
- C) -2
- D) 8

Question 145

Skill: Systems of Equations - Number of Solutions | Difficulty: Hard

The system $4x - 6y = 11$ and $kx - 18y = 34$ has no solution. What is k ?

What is k ?

- A) 4
- B) 12
- C) -12
- D) 3

Question 146

Skill: Polynomial Operations | Difficulty: Hard

A monic polynomial has zeros 2 with multiplicity 2, 5, and -3. If the polynomial is $x^4 + ax^3 + bx^2 + cx + d$, what is a ?

Student-Produced Response

Enter your answer in the response field.

Question 147

Skill: Volume & Surface Area | Difficulty: Hard

A solid sphere of radius 3 is completely submerged in a vertical cylindrical tank of radius 6. Assuming no water spills, by how many units does the water level rise?

Student-Produced Response

Enter your answer in the response field.